

Zepto E-Commerce Inventory and Product Analysis

1. Project Overview

This project analyses Zepto's e-commerce product inventory dataset to understand product pricing, discount strategies, inventory availability, and category performance using SQL and Power BI. The objective is to uncover business insights that help optimize inventory management, pricing decisions, and revenue generation through data-driven analysis.

2. Dataset Summary

- **Rows:** 3732
- **Columns:** 9 (before cleaning and feature engineering)
- **Key Features:** Product information (SKU ID, Product Name, Category), Pricing details (MRP, Discount Percentage, Discounted Selling Price), Inventory information (Available Quantity, Out of Stock), Product Weight, and Package Quantity.
- **Missing Data:** No missing values after data cleaning.

3. Exploratory Data Analysis using SQL

Exploratory Data Analysis (EDA) was performed using SQL to understand the structure and characteristics of the zepto dataset.

- Imported the dataset into PostgreSQL.
- Counted the total number of product records in the dataset.
- Checked a sample of data to understand how the data is stored.
- Examined the dataset structure and verified column data types.
- Checked for null values across all columns.
- Identified **different product categories** available in the inventory.
- Compared the number of **in-stock and out-of-stock products** to understand inventory availability.
- Identified **product names appearing multiple times**, representing different SKUs, package sizes, or product variants.
- Reviewed product pricing and inventory fields to detect inconsistencies before data cleaning.

Data cleaning was performed to improve data quality before analysis.

- Removed records with zero MRP or zero selling price.

- Converted **MRP** and **Discounted Selling Price** from paise to rupees for better readability and consistency.
- Validated inventory and pricing data after cleaning.

4. Data Analysis using SQL (Business Transactions)

Business-focused SQL queries were written to answer key inventory, pricing, and product performance questions.

- **Top 10 products with the highest discounts** – Identified products offering the best value based on discount percentage.

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Mas...	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
7	RRO Mozzarella Pizza Cheese	275.00	50.00
8	Ceres Foods Fish Mustard Instant Liquid M...	220.00	50.00
9	Dukes Waffy Strawberry Roll	150.00	50.00
10	Ceres Foods Nalli Nihari Instant Liquid Mas...	220.00	50.00

- **High-priced out-of-stock products** – Identified premium products (MRP > ₹300) that are currently unavailable, highlighting potential revenue loss.

	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
3	Aashirvaad Atta With Mutigrains	315.00
4	Everest Kashmiri Lal Chilli Powder	310.00

- **Estimated revenue by category** – Calculated the potential revenue for each product category using selling price and available inventory.

	category character varying (120) 🔒	total_revenue numeric 🔒
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00
7	Home & Cleaning	122661.00
8	Ice Cream & Desserts	224385.00
9	Chocolates & Candies	224385.00
10	Packaged Food	224385.00
11	Personal Care	270849.00
12	Paan Corner	270849.00
13	Munchies	337369.00
14	Cooking Essentials	337369.00

- **Premium products with low discounts** – Identified products priced above ₹500 with discounts below 10%, indicating premium pricing strategies.

	name character varying (150) 🔒	mrp numeric (8,2) 🔒	discountpercent numeric (5,2) 🔒
1	Dhara Kachi Ghani Mustard Oil Jar	1250.00	8.00
2	Saffola Gold (Jar)	1240.00	0.00
3	Dhara Filtered Groundnut Oil (Jar)	1050.00	1.00
4	Fortune Rice Bran Health Oil (Jar)	1050.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	1050.00	0.00
6	Fortune Soyabean Oil	1005.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	925.00	0.00
8	Surf Excel Matic Powder Front Load	810.00	7.00
9	Surf Excel Matic Top Load	720.00	9.00
10	Pedigree Puppy Dry Dog Food Food Chicken & Milk	690.00	6.00
11	Pedigree Dog Food Adult Meat & Rice	660.00	7.00
12	Lizol Double Concentrate Disinfectant Floor Cleaner Citrus	650.00	8.00
13	Nestle Nestle Nan Pro Follow-Up Formula	650.00	0.00
14	Dove Daily Shine Shampoo	645.00	5.00
15	L'Oreal Paris Excellence Creme Hair Color, 1 Black	620.00	0.00

16	L'Oreal Paris Excellence Creme Hair Color, 3 Dark Brown/Natural Darkest Br...	620.00	0.00
17	L'Oreal Paris Excellence Creme Hair Color, 4 Natural Brown/Natural Dark Br...	620.00	0.00
18	L'Oreal Paris Excellence Creme Hair Color, 4.25 Aishwarya's Brown	620.00	0.00
19	Nestle Nan Pro 4 Follow Up Formula Powder For Older Infants	620.00	0.00
20	Pediasure Premium Chocolate Powdered Health Drink	610.00	0.00
21	Surf Excel Quick Wash	600.00	8.00
22	Gowardhan Ghee Jar	598.00	3.00
23	Organic India Apple Cider Vinegar	595.00	9.00
24	Bagrry's Fruit Fibre Muesli Mixed Fruit	595.00	0.00
25	Bagrrys Fruit N Fibre Muesli - Strawberry	595.00	0.00
26	Patanjali Cow's Ghee	565.00	0.00
27	Sunsilk Lusciously Thick & Long Shampoo	555.00	6.00
28	Sunsilk Stunning Black Shine Shampoo	555.00	6.00
29	Whisper Ultra Soft XL+ Sanitary Pads	555.00	0.00
30	Kellogg's Real Almond & Honey Corn Flakes	550.00	5.00
31	L'Oreal Casting Creme Ebony Black Hair Color	550.00	0.00
32	L'Oreal Paris Casting Creme Gloss Hair Color (Dark Brown 400)	550.00	0.00
33	L'Oreal Paris Casting Creme Gloss Hair Color, Medium Brown 500	550.00	0.00
34	L'Oreal Paris Excellence Creme Hair Color, 3.16 Burgundy	550.00	0.00
35	L'Oreal Paris Shine Masque Royal Jelly Hair Colour	550.00	0.00
36	Surf Excel Matic Front Load Detergent Powder	530.00	4.00
37	Amul Ghee (Tin)	520.00	2.00
38	Fresh Mutton Mince - Kheema (Halal)	515.00	3.00
39	Amul Pure Ghee (Pouch)	505.00	1.00

- **Top 5 categories by average discount** – Determined which product categories offer the highest average discounts to customers.

	category character varying (120) 🔒	avg_discount numeric 🔒
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

- **Price per gram analysis** – Calculated the cost per gram for products weighing more than 100 grams to identify the best value-for-money products.

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)
1	Onion	3000	57.00
2	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00
3	Aashirvaad Iodised Salt	1000	19.00
4	Vicks Cough Drops Menthol	1160	20.00
5	Onion	1000	21.00
6	Tata Salt	1000	24.00
7	Potato	1000	29.00
8	Shubh kart - Tejas Twisted Cotton Wicks 1000n	1000	28.00
9	Carrot	500	15.00
10	Baby Potato	500	16.00

(more than 1000 rows)

- **Product segmentation by weight** – Categorized products into **Low**, **Medium**, and **Bulk** groups based on product weight.

	name character varying (150)	weightingms integer	weight_catgeg text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low
8	Britannia Treart Creme Wafers Chocolate	60	Low
9	Popular Essentials Dhania (Coriander Seeds)	200	Low
10	Popular Essentials Californian Almond	250	Low

(more than 1000 rows)

- **Total inventory weight by category** – Measured the total inventory weight available for each product category.

	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Ice Cream & Desserts	490797
11	Chocolates & Candies	490797
12	Packaged Food	490797
13	Munchies	1404654
14	Cooking Essentials	1404654

- **Out-of-stock percentage by category** – Calculated the proportion of unavailable products across each category to evaluate inventory performance.

	category character varying (120) 🔒	total_products bigint 🔒	outofstock_products bigint 🔒	outofstock_percent numeric 🔒
1	Personal Care	344	21	6.10
2	Paan Corner	344	21	6.10
3	Fruits & Vegetables	93	6	6.45
4	Home & Cleaning	193	19	9.84
5	Packaged Food	388	45	11.60
6	Ice Cream & Desserts	388	45	11.60
7	Chocolates & Candies	388	45	11.60
8	Munchies	514	64	12.45
9	Cooking Essentials	514	64	12.45
10	Health & Hygiene	97	13	13.40
11	Meats, Fish & Eggs	63	12	19.05
12	Dairy, Bread & Batter	129	28	21.71
13	Beverages	129	28	21.71
14	Biscuits	147	42	28.57

- **Category-wise price reduction analysis** – Measured the total price reduction offered by each category based on the difference between MRP and selling price.

	category character varying (120) 🔒	total_price_reduction numeric 🔒
1	Cooking Essentials	8074.00
2	Munchies	8074.00
3	Personal Care	5979.00
4	Paan Corner	5979.00
5	Ice Cream & Desserts	5684.00
6	Chocolates & Candies	5684.00
7	Packaged Food	5684.00
8	Home & Cleaning	2077.00
9	Dairy, Bread & Batter	1932.00
10	Beverages	1932.00
11	Meats, Fish & Eggs	1461.00
12	Health & Hygiene	1292.00
13	Biscuits	986.40
14	Fruits & Vegetables	667.00

- **Product count by category** – Counted the number of products available in each category to understand product distribution.

	category character varying (120) 🔒	no_of_products bigint 🔒
1	Meats, Fish & Eggs	63
2	Fruits & Vegetables	93
3	Health & Hygiene	97
4	Dairy, Bread & Batter	129
5	Beverages	129
6	Biscuits	147
7	Home & Cleaning	193
8	Paan Corner	344
9	Personal Care	344
10	Ice Cream & Desserts	388
11	Chocolates & Candies	388
12	Packaged Food	388
13	Munchies	514
14	Cooking Essentials	514

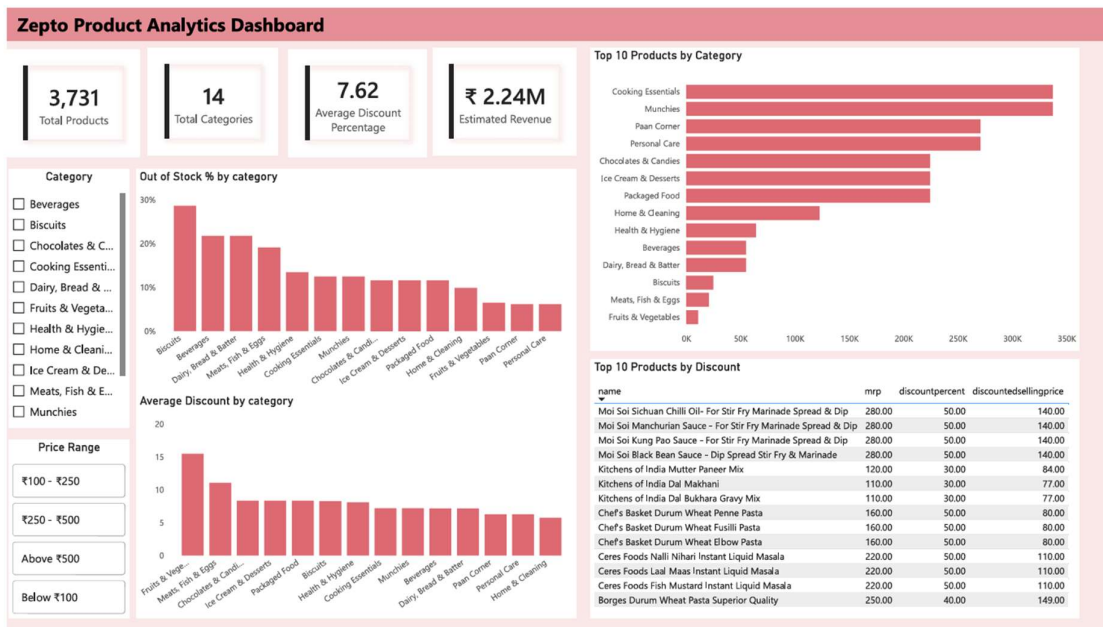
5. Dashboard in Power BI

An interactive Power BI dashboard was developed to visualize key inventory, pricing, and category performance metrics within the Zepto product dataset. The dashboard provides a comprehensive overview of product availability, discount strategies, and estimated revenue through interactive visualizations.

The dashboard includes:

- **Key Performance Indicators (KPIs):**
 - Total Products
 - Total Categories
 - Average Discount Percentage
 - Estimated Revenue
- **Category-wise Product Distribution** – Displays the top product categories based on the number of available products.
- **Out-of-Stock Percentage by Category** – Highlights categories with the highest proportion of unavailable products, helping identify inventory shortages.
- **Average Discount by Category** – Compares discount percentages across different product categories to evaluate promotional strategies.
- **Top 10 Products by Discount** – Lists products offering the highest discounts along with their MRP and discounted selling prices.
- **Interactive Filters** – Allows users to analyse data by **Category** and **Price Range** for more detailed insights.

The dashboard enables users to monitor inventory availability, compare pricing strategies, identify high-performing categories, and support data-driven decisions for inventory management and product pricing.



6. Key Insights

- Several products offer discounts of 50% or more, indicating aggressive promotional strategies to attract customers and increase product visibility.
- Some high-priced products are currently out of stock, suggesting potential missed revenue opportunities if customer demand remains high.
- A number of products are sold without any discount, indicating strong customer demand or premium positioning that does not require promotional pricing.
- Fruits & Vegetables have the highest average discount (approximately 15%), showing that this category receives the most frequent price reductions.
- Price-per-gram analysis helps identify products that provide better value for money and reveals pricing differences across similar products.
- Products are distributed across Low, Medium, and Bulk weight categories, providing insights into inventory composition and product packaging.
- Total inventory weight varies significantly across categories, highlighting differences in storage and warehouse requirements.
- Out-of-stock percentage analysis identifies categories with frequent stock shortages, enabling better inventory monitoring.
- Category-wise price reduction analysis reveals which categories receive the largest overall discounts, helping evaluate promotional spending.
- Product distribution analysis shows how products are spread across different categories, providing insights into assortment diversity and inventory allocation.

7. Business Recommendations

- **Review promotional campaigns** for heavily discounted products to ensure increased sales justify the reduced profit margins.

- **Prioritize restocking high-value products** that are frequently out of stock to minimize potential revenue loss and improve customer satisfaction.
- **Maintain premium pricing** for products that continue to sell well without discounts, avoiding unnecessary price reductions.
- **Evaluate discount strategies** in the Fruits & Vegetables category to balance customer attraction with overall profitability.
- **Leverage price-per-gram insights** to improve pricing consistency and promote value-for-money products to customers.
- **Use weight-based product segmentation** to optimize packaging, warehouse organization, delivery planning, and bulk order management.
- **Monitor total inventory weight** to improve warehouse space utilization and logistics planning.
- **Track out-of-stock trends regularly** to improve inventory replenishment and reduce stock shortages.
- **Analyze category-wise price reductions** to optimize promotional budgets and identify categories where discounts are most effective.
- **Review product distribution across categories** to maintain a balanced product portfolio and identify opportunities to expand underrepresented categories.

8. Conclusion

This project demonstrates how SQL and Power BI can be used to analyse an e-commerce product inventory dataset and generate actionable business insights. The findings help understand pricing strategies, inventory availability, category performance, and revenue opportunities, enabling better business decisions in a retail and quick-commerce environment.